

Complete DevOps A-Z Notes (2025 Edition)

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1. SDLC (Software Development Life Cycle)

- Definition: SDLC defines structured stages in software development: Requirement, Design, Implementation, Testing, Deployment, Maintenance.
- Models: Waterfall, Agile, V-Model, Spiral, Iterative.

2. Agile

- Iterative, customer-centric methodology.
- Frameworks: Scrum, Kanban, XP, SAFe.
- Agile Manifesto: 12 principles emphasizing customer collaboration and adaptability.

3. Scrum

- Roles: Product Owner, Scrum Master, Development Team.
- Artifacts: Product Backlog, Sprint Backlog, Increment.
- Ceremonies: Sprint Planning, Daily Stand-up, Review, Retrospective.

4. Git & GitHub

- Git: Version Control System (track, manage code).
- GitHub: Remote platform for Git repositories.
- Commands: git init, add, commit, push, pull, branch, merge.

5. CI/CD

- Continuous Integration: frequent code merging, auto-testing.
- Continuous Delivery: automated release pipeline.
- Continuous Deployment: automatic production release.

6. Jenkins

- Automation server for CI/CD.
- Jenkinsfile defines pipeline stages.

Example:

```
pipeline {  
  agent any  
  stages {  
    stage('Build'){steps{echo 'Building...'}}  
    stage('Test'){steps{echo 'Testing...'}}  
    stage('Deploy'){steps{echo 'Deploying...'}}  
  }  
}
```

7. YAML

- Configuration format used in Jenkins, K8s, Ansible.

Example:

```
version: '3'  
services:  
  web:  
    image: nginx  
    ports: ["80:80"]
```

8. Docker

- Containerization platform.
- Components: Image, Container, Dockerfile, Docker Hub.

Dockerfile Example:

```
FROM python:3.9  
WORKDIR /app  
COPY . .  
RUN pip install -r requirements.txt  
CMD ["python", "app.py"]
```

9. Kubernetes

- Container orchestration tool.
- Components: Pod, Node, Deployment, Service, ConfigMap, Secret.

Example Deployment YAML:

apiVersion: apps/v1

kind: Deployment

metadata:

name: myapp

spec:

replicas: 3

selector:

matchLabels:

app: myapp

template:

metadata:

labels:

app: myapp

spec:

containers:

- name: myapp

image: nginx

ports:

- containerPort: 80

10. Terraform & IaC

- Infrastructure as Code tool (HCL language).

Example:

```
provider "aws" {  
  region = "ap-south-1"  
}  
  
resource "aws_instance" "web" {  
  ami = "ami-0abcdef1234567890"  
  instance_type = "t2.micro"  
}
```

11. Cloud Basics (AWS & Azure)

- IaaS, PaaS, SaaS.

- AWS: EC2, S3, RDS, Lambda, VPC.
- Azure: VM, Blob Storage, Functions, SQL DB.

12. Monitoring & DevSecOps

- Tools: Prometheus, Grafana, ELK, Nagios.
- DevSecOps: integrate security in CI/CD.

Practices: code scanning, secret management, access control.

DevOps Flow:

Code -> GitHub -> Jenkins -> Docker -> Kubernetes -> Terraform -> Cloud -> Monitor.